ClipSAT : Cambridge & Edexcel Additional Mathematics 9606/4PM1 [ANSWER KEY]

Unit 1.01: Functions, Modulus & Quadratics

Q1. The function fx = frac{3x + 1}{x - 2} is defined for x ne 2 . Find the inverse function f^{-1}x .

[ ] A: f^{-1}x = frac{3x - 2}{x + 1}
[ ] B: f^{-1}x = frac{2x - 1}{x - 3}
[ ] C: f^{-1}x = frac{x - 2}{3x + 1}
[x] D: f^{-1}x = frac{2x + 1}{x - 3} quad x

Solution: Let y = frac{3x + 1}{x - 2} implies yx - 2 = 3x + 1 implies yx - 2y = 3x + 1 implies xy - 3 = 2y + 1 implies x = frac{2y + 1}{y - 3} . Swap variables to obtain f^{-1}x = frac{2x + 1}{x - 3} . [M1][A1]

Q2. The function fx = frac{4x + 1}{x - 2} is defined for x ne 2 . Find the inverse function f^{-1}x .

[ ] A: f^{-1}x = frac{2x - 1}{x - 4}
[ ] B: f^{-1}x = frac{x - 2}{4x + 1}
[x] C: f^{-1}x = frac{2x + 1}{x - 4} quad x
[ ] D: f^{-1}x = frac{4x - 2}{x + 1}

Solution: Let y = frac{4x + 1}{x - 2} implies yx - 2 = 4x + 1 implies yx - 2y = 4x + 1 implies xy - 4 = 2y + 1 implies x = frac{2y + 1}{y - 4} . Swap variables to obtain f^{-1}x = frac{2x + 1}{x - 4} . [M1][A1]

Q3. The function fx = frac{5x + 1}{x - 2} is defined for x ne 2 . Find the inverse function f^{-1}x .

[x] A: f^{-1}x = frac{2x + 1}{x - 5} quad x
[ ] B: f^{-1}x = frac{x - 2}{5x + 1}
[ ] C: f^{-1}x = frac{5x - 2}{x + 1}
[ ] D: f^{-1}x = frac{2x - 1}{x - 5}

Solution: Let y = frac{5x + 1}{x - 2} implies yx - 2 = 5x + 1 implies yx - 2y = 5x + 1 implies xy - 5 = 2y + 1 implies x = frac{2y + 1}{y - 5} . Swap variables to obtain f^{-1}x = frac{2x + 1}{x - 5} . [M1][A1]

Q4. The function fx = frac{6x + 1}{x - 2} is defined for x ne 2 . Find the inverse function f^{-1}x .

[x] A: f^{-1}x = frac{2x + 1}{x - 6} quad x
[ ] B: f^{-1}x = frac{x - 2}{6x + 1}
[ ] C: f^{-1}x = frac{2x - 1}{x - 6}
[ ] D: f^{-1}x = frac{6x - 2}{x + 1}

Solution: Let y = frac{6x + 1}{x - 2} implies yx - 2 = 6x + 1 implies yx - 2y = 6x + 1 implies xy - 6 = 2y + 1 implies x = frac{2y + 1}{y - 6} . Swap variables to obtain f^{-1}x = frac{2x + 1}{x - 6} . [M1][A1]

Q5. The function fx = frac{7x + 1}{x - 2} is defined for x ne 2 . Find the inverse function f^{-1}x .

[ ] A: f^{-1}x = frac{7x - 2}{x + 1}
[ ] B: f^{-1}x = frac{2x - 1}{x - 7}
[ ] C: f^{-1}x = frac{x - 2}{7x + 1}
[x] D: f^{-1}x = frac{2x + 1}{x - 7} quad x

Solution: Let y = frac{7x + 1}{x - 2} implies yx - 2 = 7x + 1 implies yx - 2y = 7x + 1 implies xy - 7 = 2y + 1 implies x = frac{2y + 1}{y - 7} . Swap variables to obtain f^{-1}x = frac{2x + 1}{x - 7} . [M1][A1]

Q6. The function $f(x) = \frac{8x + 1}{x - 2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

[x] A: $f^{-1}(x) = \frac{2x + 1}{x - 8} \quad x$

[ ] B: $f^{-1}(x) = \frac{x - 2}{8x + 1}$

[ ] C: $f^{-1}(x) = \frac{2x - 1}{x - 8}$

[ ] D: $f^{-1}(x) = \frac{8x - 2}{x + 1}$

Solution: Let $y = \frac{8x + 1}{x - 2} \implies yx - 2 = 8x + 1 \implies yx - 2y = 8x + 1 \implies xy - 8 = 2y + 1 \implies x = \frac{2y + 1}{y - 8}$. Swap variables to obtain $f^{-1}(x) = \frac{2x + 1}{x - 8}$. [M1][A1]

Q7. The function $f(x) = \frac{9x + 1}{x - 2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

[ ] A: $f^{-1}(x) = \frac{2x - 1}{x - 9}$

[x] B: $f^{-1}(x) = \frac{2x + 1}{x - 9} \quad x$

[ ] C: $f^{-1}(x) = \frac{x - 2}{9x + 1}$

[ ] D: $f^{-1}(x) = \frac{9x - 2}{x + 1}$

Solution: Let $y = \frac{9x + 1}{x - 2} \implies yx - 2 = 9x + 1 \implies yx - 2y = 9x + 1 \implies xy - 9 = 2y + 1 \implies x = \frac{2y + 1}{y - 9}$. Swap variables to obtain $f^{-1}(x) = \frac{2x + 1}{x - 9}$. [M1][A1]